

Name Of The Student : NISHI GANDHA RAI

Student Roll No : PG190628

Name Of The Program : Master's in Architectural Design

Course : AD4009 : Between Campus and Commons-
Situation: Indian Institute of Management,
Ahmedabad

Course Tutor/s :



Shubhra Rajee



Kevin Mark Low



Link to CEPT Portfolio :

(studio work from past semesters)

Course Expense :

(INR -extra expenditure a
student would incur during
the semester)

Rs. 5,000 (+/-) (FIVE THOUSAND Rupees)

Long Description :

The academic campus as a hallowed precinct distinct from its surroundings has persisted within our architectural imagination since education began, when the first centers of knowledge were monasteries and gurukuls. Privileging the entitled, they made knowledge sacred and gave birth to the paradigm which maintains that theoretical excellence can only come through freedom from the practical mundane and its constraints. The studio suggests that the IIM Ahmedabad is no different in this regard and reconsiders this distancing. By analyzing the iconic campus through a systematic inquiry into its configuration, programming, and consequence, we first identified problems inherent in the existing design by way of vital relationships it failed to consider. Subtle, yet profound intervention/s that will resolve the problems identified were then proposed by the students, in a simultaneous attempt to forge relevant new relationships with the neighborhoods that surround it, to address the urban island that the campus currently is. Shifting the emphasis from design solutions derived primarily from the aesthetic expression of form and volume to that of solving real problems, students rekindled the potential of programme, the meaning of use, and continuity of the built environment, as an understanding of what constitutes critical architecture in service of responsibility over mere reputation

Course Prerequisite :

For MAAD Semester 3 Students only

Problem Statement :

How do we go beyond personal visions of beauty or the future to elevate our understanding of architecture from merely beautiful buildings and reputation to one of profound relationships in service of responsibility?

Learning Outcomes :

After completing the Studio, the student will be able to :

- Analyze and articulate - through a drawing-based methodology - the iconic and the everyday within built environments, and the relationships that bind them (site + context analysis)
- Develop a systematic inquiry into the conditions and consequences of precedents in order to articulate their limitations and viability, with an emphasis on modern architecture (precedent/ case study analysis)
- Build up to a coordinated physical solution, finding development through the lens of profound relationships, rather than that of purely expressive shape and space-making (programming & design development)
- Use drawings/ models as tools for thinking, rather than merely for representation (finding relevance for skills)

Weekly Plan & Assessment:

Weeks	Exercise Description	Assessment	
		%	Criteria
Week 1	Site Analysis (Jul 27-31)	0	n/a
Week 2	Site Analysis (Aug 3-7)	0	n/a
Week 3	Design Hypothesis (Aug 10-14)	10	Review 01 w Internal/ External Critics (TBD) - Aug 14/15, 2020
Week 4	Situating the Hypothesis: Design Phase 1 (Aug 17-21)	0	n/a
Week 5	Situating the Hypothesis: Design Phase 1 (Aug 24-28)	0	n/a
Week 6	Situating the Hypothesis: Design Phase 1 (Aug 31-Sept 4)	10	Review 02 w External Critics - Sept 4/5, 2020
Week 7	Workshop: Detail 1 (Sept 7-11)	0	n/a
Week 8	Design Phase 2 (Sept 14-18)	0	n/a
Week 9	Design Phase 2 (Sept 21-25)	0	n/a
Week 10	Design Phase 2 (Sept 28-Oct 2)	10	Review 03 w External Critics - Oct 2/3, 2020 (mid-sem grades out of 30)
Week 11	Workshop: Detail 2 / Design Phase 3 (TBD) (Oct 5-9)	0	n/a
Week 12	Design Phase 3 (Oct 12-16)	0	n/a
Week 13	Design Phase 3 (Oct 19-23)	0	n/a
Week 14	Design Phase 3 (Oct 26-30)	0	n/a
Week 15	Re-membering the Site (Nov 2-6)	0	n/a
Week 16	Synthesis (Nov 9-12)	40	Review 04 (final) w External Critics - Nov 12/13, 2020

REFERENCE :

All readings will be distributed during the semester, specific to course exercises and assignments.

Co-Tutor Profile :

'Shubhra's practice involves designing buildings and teaching architecture. With graduate and postgraduate degrees in architecture and a minor in theory and criticism from Cornell and CEPT Universities, she founded shubhra raje_built environments, an architecture and design studio that concerns itself with relevant design, architectural economy and spatial ecology.'

Co-Tutor Profile :

'Kevin has taught and worked in the monsoon tropics since graduating from the United States with graduate and postgraduate degrees in architecture, and a minor in art history. His work and writing has been published widely, directed through an understanding that the big idea is less about the search for radical solutions, than about the discovery of radical questions relevant to specific context.'

Course :

AD4603 : Concern, Context, Concept: Theory and Design relationships

Course Tutor/s :



Kaiwan Mehta



Link to CEPT Portfolio :

(studio work from past semesters)

Course Expense :

(INR -extra expenditure a student would incur during the semester)

2000/- (TWO THOUSAND Rupees)

Long Description :

This seminar will introduce and equip students to engage with theoretical frameworks around key issues/concepts that are of contemporary relevance to the field of architecture globally and developing an approach to one's own practice in India. A set of introductory lecture-discussions as well as readings will introduce students to the range of design philosophies and concepts we engage with, in the process of designing and shaping architecture in the contemporary context. The seminar course proposes that the broad range of our choices, approaches, and intentions in design could be categorised into the following three: (1) an area concern or (2) context or (3) a concept. It is through one of these entry points that one can develop a design strategy as well as a direction for an architectural practice. On the basis of the discussions and areas of concern, or context, or concept, identified by a student s/he will be encouraged to research writings, arguments, debates available globally as well as conduct a series of case studies of buildings/projects that deal with the same. A process of analysis and synthesis towards developing a design approach in architecture practice will be encouraged from these exercises of textual and building case studies.

Course Prerequisite :

Mandatory for MAD 3rd semester students and open as an elective for MAHT, MCR and MLA students only.

Problem Statement :

Learning Outcomes :

After completing the Courses, the student will be able to :

- Find, examine, and illustrate the areas of architectural issues through which one wishes to approach design process and studio practice
- Classify design processes and categorise issues that shape and concern a design practice in the contemporary world
- Plan and design an architectural approach for a building or a project or a studio

Course Structure :

The course begins with exploring and discussing collectively the range of issues that influence architectural design and process in the contemporary world. These will be substantiated with readings and arguments available in architecture publications, lectures, interviews and other forums of discourse.

Discuss with examples and situations at hand how architectural design approaches can be categorised into Concern, Context, and Concept. Through case studies one will further develop a structure to inspect and analyse, as well as learn about processes of design.

The seminar will then discuss how theoretical concerns as well as architectural projects can be reviewed to nourish one's own design approach and intent.

REFERENCE :

Architecture Theory since 1968 edited by Michael K Hays

Constructing a New Agenda: Architectural Agenda 1993-2009 edited by Krista Sykes

Tutor Profile :

'A theorist and critic in the fields of visual culture, architecture, and city studies, he has studied Architecture, Literature, Indian Aesthetics and Cultural Studies. He is the Managing Editor of Domus India and Professor and Programme Chair of the Doctoral Programme, FA, CEPT. He was the Charles Correa Chair Professor for the academic year 2017-2018. He authored *Alice in Bhuleshwar: Navigating a Mumbai Neighbourhood* (Yoda Press. New Delhi, 2009) and *The Architecture of I M Kadri* (Niyogi. New Delhi, 2016). He has co-curated two national exhibitions on Architecture as well as Housing in India, besides other exhibitions curated independently; he has authored a section on India and South-Asia since 1947 to the updated edition of Bannister Fletcher's "A History of World Architecture"

Course : AD4604 : Contemporary Architecture: Practices and Processes in India

Course Tutor/s :



Kaiwan Mehta

Link to CEPT Portfolio :
(studio work from past semesters)

Course Expense : 2000/- (TWO THOUSAND Rupees)
(INR -extra expenditure a student would incur during the semester)



Long Description :

The course will survey as well as understand and debate the practice of architecture in contemporary India. A study and understanding of the various factors that influence and shape architectural ideas and design decisions in India will be an important component of this seminar. The seminar begins with the question - how does an architect in India today make choices as a designer, as a technocrat, as a professional, as a thinker, and what would be her/his areas of influence and impact? Recent developments in India - in spheres of urbanisation, technology, politics, and culture that shape the current 'state of architecture' in India since the 1990s will be discussed. Key practices and projects will be taken up for study and debate, and reviewed through popular and theoretical arguments that analyse and discuss the contemporary moment in architecture in India. The ecosystem within which architecture functions and is shaped will be explored, with a focus on the ideas of design we pursue, as well as debating the 'role of the architect'. Architecture's existence between fashion and lifestyle, to social engagement and political sustenance will be a governing concern of this course.

Course Prerequisite : Mandatory for MAD 1st semester students; and open as an elective for students from in 4th or 5th year of B. Arch programme.

Problem Statement :

Learning Outcomes : After completing the Courses, the student will be able to :

- Describe, explain, and compare architectural projects in the contemporary for their design ideas as well as production in India since 1990s
- Examine, explain, and investigate the issues and influences an architectural practice in India faces and encounters today
- Explain and outline key texts written on architecture in contemporary India
- Construct and synthesise theoretical arguments into certain aspects of a design approach and process

Course Structure :

The course will identify 8-10 key texts, and a basket of about 20-25 projects to undertake a study of contemporary architecture in India, in the context of current global developments. The course will consider 1989 and 1991-92 as watershed dates to understand the idea of the contemporary in world and in India, respectively.

The texts will primarily be from the world of architecture writings such as project reviews, critical essays, awards jury reports, etc. but also include some reference texts from various other fields such as cultural theory, visual theory, semiotics, economics, etc.

Ecology of architectural practice in India will be crucial initial step; after which case studies of buildings and architectural projects will be undertaken. Methods of case study and analysis will be discussed and provided in the seminar through an curated list of samples focussing on methodologies of approaching analysis and critical reviews.

The seminar is developed keeping in mind the world of operations available to a practicing architect in India today.

Keeping in mind contemporary trends as well as tendencies that occupy the world of the architect today, discussions and readings or case studies will be proposed in every class.

All of the above will finally lead to evaluating and debating the present status of the architect in society as well the scope and limitation of his profession and zone of work and action. This discussion will also involve how design decisions, architectural language, and the building idea are linked with the imagination and perception of the role of the architect, and her/his standing within the public and professional sphere.

REFERENCE :

The course will draw on essays and books by: Romi Khosla, Gautam Bhatia, Rahul Mehrotra, Ranjit Hoskote, A Srivathsan, Peter Scriver, Amit Srivastava, Madhavi Desai, Mary Woods, Jon Lang, Kaiwan Mehta, Rem Koolhaas, Allan de Botton, Hal Foster, Anthony Vidler, and others.

Co-Tutor Profile :

'A theorist and critic in the fields of visual culture, architecture, and city studies, he has studied Architecture, Literature, Indian Aesthetics and Cultural Studies. He is the Managing Editor of Domus India and Professor and Programme Chair of the Doctoral Programme, FA, CEPT. He was the Charles Correa Chair Professor for the academic year 2017-2018. He authored *Alice in Bhuleshwar: Navigating a Mumbai Neighbourhood* (Yoda Press, New Delhi, 2009) and *The Architecture of I M Kadri* (Niyogi, New Delhi, 2016). He has co-curated two national exhibitions on Architecture as well as Housing in India, besides other exhibitions curated independently; he has authored a section on India and South-Asia since 1947 to the updated edition of Bannister Fletcher's "A History of World Architecture"

Course : AD4607 : Computational Design and Digital Fabrication

Course Tutor/s :



Shehzad Irani

Link to CEPT Portfolio :
(studio work from past semesters)

Course Expense : 25000/- (TWENTY FIVE THOUSAND Rupees)
(INR -extra expenditure a student would incur during the semester)



Long Description :

This is an introductory course to computational design and the prerequisite for a monsoon semester. This course is primarily intended for students with little background in CAD applications / programming and who are interested in developing their skills in order to be able to better understand, interface with and customize the digital tools they are using, or develop their own software and interactive applications. The course deals with basics and more advanced topics in the field. The course introduces students to fundamental concepts and techniques in computational design as well as the relevant mathematics. By the term “computational design” we mean an ad hoc set of methods borrowed from computer science, computational geometry and other fields, and adapted to specific design problems such as design development, fabrication, analysis, interaction and communication. Students will be learning the basics of software like rhino and grasshopper as also using digital production techniques like 3d printing, CNC plotting, cutting, etching etc. At the end of the course the students should be able to write a basic routine, engage with mathematics based form generation and materializing their designs through 3d printing, CNC cutting/plotting or routing.

Course Prerequisite : Mandatory for Semester 3 MAD students.

Problem Statement :

Learning Outcomes :

After completing the Courses, the student will be able to :

- Understand the difference between conventional CAD based design-simulation and parametric design-simulation-optimization.
- To interface and customize digital tools to improve workflow in design, analysis and geometry generation and general optimization.
- To compile, use and apply methods of coding to form-making and form finding by using digital tools and techniques.
- To be able to create and analyze emerging forms using latest digital tools. To create components, elements and parts to create a whole using computational tools
- To familiarize and apply software like Rhino, Grasshopper, scripting and other tools for the same. To equip students to be able to use, manipulate and apply computer-controlled hardware like 3d printers, routers, laser cutters etc to create, represent and develop their designs.

REFERENCE :

Elements of parametric design by Woodbury, Robert.
From control to design parametric/algorithmic architecture by Sakamoto, Tomoko & Others
Contemporary architecture and the digital design process by Szalapaj, Peter
Finite element analysis by Narasaiah, Lakshmi G.
Parametric architecture with grasshopper primer by Tedeschi, Arturo
Computational design thinking by Menges, Achim Ed.
Shell structures for architecture : by Sigrid Adriaenssens & others. Emergent technologies and design :

by Jensen, Michael & others

New structuralism : design, engineering and architectural technologies by Oxman, Rivka

Ed.

Form follows nature : a history of nature as model for design in engineering, architecture and art. Tr.

English-German by Jeremy Gains, Rudolf Finsterwalder & Johnathan Buckley by Finsterwalder, Rudolf

Ed.

Structural basis of architecture. by Sandaker, Bjorn N. & others

Co-Tutor Profile :

'Shehzad Irani is an alumni of SBST and HS-Anhalt and has been a professor at CEPT university since 2006. He has guided numerous theses at CEPT. He has also co-authored papers in Concrete and structure, which have been published internationally. He is the founder-operator of SCHAFBOCK DESIGN+WORKSHOP a firm concentrating on tensile structures and architectural engineering.'